

PSet5 Solutions

Problem 1: Find First Partial Derivatives

(a)

$$f(x, y) = \frac{1}{3}x^5 + 3x^3y^2 + 3xy^4.$$

Taking the partial derivative with respect to x ,

$$f_x(x, y) = \frac{5}{3}x^4 + 9x^2y^2 + 3y^4$$

Taking the partial derivative with respect to y ,

$$f_y(x, y) = 6x^3y + 12xy^3$$

(b)

$$g(x, y) = 4xe^{4y+1}.$$

With respect to x ,

$$g_x(x, y) = 4e^{4y+1}$$

With respect to y ,

$$g_y(x, y) = 4x \left(4e^{4y+1} \right),$$

so

$$g_y(x, y) = 16xe^{4y+1}.$$

(c)

$$k(x, y) = \frac{x - y}{x + y}.$$

Using the quotient rule,

$$k_x = \frac{(x+y) - (x-y)}{(x+y)^2} = \boxed{\frac{2y}{(x+y)^2}}.$$

Similarly,

$$k_y = \frac{-(x+y) - (x-y)}{(x+y)^2} = \boxed{-\frac{2x}{(x+y)^2}}.$$

Problem 2: Gradient, Jacobian, and Hessian

(a)

$$g(x, y) = x^4 + y^4 - 4xy + 2.$$

i. Gradient

The first partial derivatives are

$$g_x = 4x^3 - 4y,$$

and

$$g_y = 4y^3 - 4x.$$

Therefore,

$$\boxed{\nabla g(x, y) = \begin{pmatrix} 4x^3 - 4y \\ 4y^3 - 4x \end{pmatrix}}$$

At $(x, y) = (1, 1)$,

$$\nabla g(1, 1) = \begin{pmatrix} 4 - 4 \\ 4 - 4 \end{pmatrix} = \boxed{\begin{pmatrix} 0 \\ 0 \end{pmatrix}}.$$

ii. Jacobian

Because g is a scalar-valued function, its Jacobian is the row vector

$$\boxed{J_g(x, y) = \begin{pmatrix} 4x^3 - 4y & 4y^3 - 4x \end{pmatrix}}.$$

iii. Hessian

The second partial derivatives are

$$g_{xx} = 12x^2, \quad g_{yy} = 12y^2,$$

and

$$g_{xy} = g_{yx} = -4.$$

Hence,

$$H_g(x, y) = \begin{pmatrix} 12x^2 & -4 \\ -4 & 12y^2 \end{pmatrix}.$$

(b)

$$k(x, y) = (1 + xy)(x + y).$$

It is useful to expand first:

$$k(x, y) = x + y + x^2y + xy^2.$$

i. Gradient

$$k_x = 1 + 2xy + y^2,$$

and

$$k_y = 1 + x^2 + 2xy.$$

Thus,

$$\nabla k(x, y) = \begin{pmatrix} 1 + 2xy + y^2 \\ 1 + x^2 + 2xy \end{pmatrix}.$$

At $(1, -1)$,

$$k_x(1, -1) = 1 + 2(1)(-1) + (-1)^2 = 0,$$

and

$$k_y(1, -1) = 1 + 1^2 + 2(1)(-1) = 0.$$

Therefore,

$$\nabla k(1, -1) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

ii. Jacobian

$$J_k(x, y) = \begin{pmatrix} 1 + 2xy + y^2 & 1 + x^2 + 2xy \end{pmatrix}.$$

iii. Hessian

$$k_{xx} = 2y,$$

$$k_{yy} = 2x,$$

and

$$k_{xy} = k_{yx} = 2x + 2y.$$

Therefore,

$$H_k(x, y) = \begin{pmatrix} 2y & 2x + 2y \\ 2x + 2y & 2x \end{pmatrix}.$$

Problem 3: Find the Critical Points

For a function of two variables, critical points satisfy

$$\nabla f(x, y) = 0.$$

For a Hessian matrix

$$H = \begin{pmatrix} A & B \\ B & C \end{pmatrix},$$

the second derivative test uses

$$D = AC - B^2.$$

If $D > 0$ and $A > 0$, the point is a local minimum. If $D > 0$ and $A < 0$, the point is a local maximum. If $D < 0$, the point is a saddle point. If $D = 0$, the test is inconclusive.

(a)

$$h(x, y) = x^4 + y^4.$$

First calculate the gradient:

$$\nabla h(x, y) = \begin{pmatrix} 4x^3 \\ 4y^3 \end{pmatrix}.$$

Set the gradient equal to zero:

$$4x^3 = 0, \quad 4y^3 = 0.$$

Therefore,

$$x = 0, \quad y = 0.$$

Thus, the only critical point is

$$\boxed{(0, 0)}.$$

The Hessian is

$$H_h(x, y) = \begin{pmatrix} 12x^2 & 0 \\ 0 & 12y^2 \end{pmatrix}.$$

At $(0, 0)$,

$$H_h(0, 0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Thus,

$$D = (0)(0) - 0^2 = 0.$$

Therefore, the Hessian test is inconclusive. However,

$$h(x, y) = x^4 + y^4 \geq 0$$

for all (x, y) , with equality only at $(0, 0)$. Hence,

$$\boxed{(0, 0) \text{ is a strict local and global minimum}}$$

with minimum value

$$\boxed{h(0, 0) = 0}.$$

There are no local maxima or saddle points.

(b)

$$g(x, y) = x^4 + y^4 - 4xy + 2.$$

The gradient is

$$\nabla g(x, y) = \begin{pmatrix} 4x^3 - 4y \\ 4y^3 - 4x \end{pmatrix}.$$

Set the gradient equal to zero:

$$4x^3 - 4y = 0, \quad 4y^3 - 4x = 0.$$

Thus,

$$y = x^3$$

and

$$x = y^3.$$

Substituting $y = x^3$ into $x = y^3$ gives

$$x = (x^3)^3 = x^9.$$

Therefore,

$$x^9 - x = 0,$$

so

$$x(x^8 - 1) = 0.$$

The real solutions are

$$x = 0, \quad x = 1, \quad x = -1.$$

Since $y = x^3$, the corresponding critical points are

$$\boxed{(0, 0), (1, 1), (-1, -1)}.$$

The Hessian is

$$H_g(x, y) = \begin{pmatrix} 12x^2 & -4 \\ -4 & 12y^2 \end{pmatrix}.$$

The determinant test gives

$$D = (12x^2)(12y^2) - (-4)^2 = 144x^2y^2 - 16.$$

At $(0, 0)$,

$$D = -16 < 0.$$

Therefore,

$$\boxed{(0, 0) \text{ is a saddle point}}.$$

At $(1, 1)$,

$$D = 144 - 16 = 128 > 0,$$

and

$$g_{xx} = 12 > 0.$$

Therefore,

$$\boxed{(1, 1) \text{ is a local minimum}}.$$

Its value is

$$g(1, 1) = 1 + 1 - 4 + 2 = \boxed{0}.$$

At $(-1, -1)$,

$$D = 128 > 0$$

and

$$g_{xx} = 12 > 0.$$

Therefore,

$$\boxed{(-1, -1) \text{ is a local minimum.}}$$

Its value is

$$g(-1, -1) = 1 + 1 - 4 + 2 = \boxed{0}.$$

Thus the local minima are

$$\boxed{(1, 1) \text{ and } (-1, -1)}$$

with minimum value

$$\boxed{0}.$$

There are no local maxima, and

$$\boxed{(0, 0)}$$

is a saddle point.

(c)

$$k(x, y) = (1 + xy)(x + y).$$

Expanding,

$$k(x, y) = x + y + x^2y + xy^2.$$

The gradient is

$$\nabla k(x, y) = \begin{pmatrix} 1 + 2xy + y^2 \\ 1 + x^2 + 2xy \end{pmatrix}.$$

Set the gradient equal to zero:

$$1 + 2xy + y^2 = 0,$$

$$1 + x^2 + 2xy = 0.$$

Subtracting the second equation from the first gives

$$y^2 - x^2 = 0.$$

Therefore,

$$(y - x)(y + x) = 0,$$

so either

$$y = x$$

or

$$y = -x.$$

If $y = x$, then

$$1 + 3x^2 = 0,$$

which has no real solutions.

If $y = -x$, then

$$1 - x^2 = 0,$$

so

$$x = \pm 1.$$

Thus the critical points are

$$\boxed{(1, -1), (-1, 1)}.$$

The Hessian is

$$H_k(x, y) = \begin{pmatrix} 2y & 2x + 2y \\ 2x + 2y & 2x \end{pmatrix}.$$

At $(1, -1)$,

$$H_k(1, -1) = \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix}.$$

Thus,

$$D = (-2)(2) - 0^2 = -4 < 0.$$

Therefore,

$$\boxed{(1, -1) \text{ is a saddle point}}.$$

At $(-1, 1)$,

$$H_k(-1, 1) = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}.$$

Thus,

$$D = (2)(-2) - 0^2 = -4 < 0.$$

Therefore,

$$\boxed{(-1, 1) \text{ is a saddle point}}.$$

Hence, k has no local minima or local maxima, and its saddle points are

$$\boxed{(1, -1) \text{ and } (-1, 1)}.$$

Problem 4: Definite and Indefinite Integrals

(a)

$$\int_{-1}^0 (3x^2 - 3) dx.$$

An antiderivative is

$$F(x) = x^3 - 3x.$$

Therefore,

$$\int_{-1}^0 (3x^2 - 3) dx = [x^3 - 3x]_{-1}^0.$$

Thus,

$$= (0^3 - 3 \cdot 0) - [(-1)^3 - 3(-1)]$$

$$= 0 - (-1 + 3) = -2.$$

Therefore,

$$\boxed{-2}.$$

(b)

$$\int_2^4 e^{3y} dy.$$

An antiderivative is

$$\frac{1}{3}e^{3y}.$$

Therefore,

$$\int_2^4 e^{3y} dy = \left[\frac{1}{3}e^{3y} \right]_2^4.$$

Thus,

$$= \frac{1}{3}e^{12} - \frac{1}{3}e^6.$$

Hence,

$$\boxed{\frac{e^{12} - e^6}{3}}.$$

(c)

$$\int_3^3 \sqrt{x^5 - 2} dx.$$

Since the upper and lower limits of integration are the same,

$$\int_a^a f(x) dx = 0.$$

Therefore,

$$\boxed{\int_3^3 \sqrt{x^5 - 2} dx = 0}.$$

(d)

$$\int (x^2 - x^{-1/2}) dx.$$

Using the power rule,

$$\int x^2 dx = \frac{x^3}{3},$$

and

$$\int x^{-1/2} dx = \frac{x^{1/2}}{1/2} = 2x^{1/2}.$$

Therefore,

$$\int (x^2 - x^{-1/2}) dx = \frac{x^3}{3} - 2x^{1/2} + C.$$

Hence,

$$\boxed{\frac{x^3}{3} - 2\sqrt{x} + C}.$$

Problem 5: Substitution and Integration by Parts

(a)

$$\int 2x(x^2 + 1)^4 dx.$$

Let

$$u = x^2 + 1.$$

Then

$$du = 2x dx.$$

Therefore,

$$\int 2x(x^2 + 1)^4 dx = \int u^4 du.$$

Thus,

$$= \frac{u^5}{5} + C.$$

Substituting back,

$$\boxed{\frac{(x^2 + 1)^5}{5} + C}.$$

(b)

$$\int \frac{3x^2}{x^3 + 5} dx.$$

Let

$$u = x^3 + 5.$$

Then

$$du = 3x^2 dx.$$

Therefore,

$$\int \frac{3x^2}{x^3 + 5} dx = \int \frac{1}{u} du.$$

Hence,

$$\boxed{\ln |x^3 + 5| + C}.$$

(c)

$$\int x e^{2x} dx.$$

Use integration by parts:

$$\int u dv = uv - \int v du.$$

Choose

$$u = x, \quad dv = e^{2x} dx.$$

Then

$$du = dx, \quad v = \frac{1}{2} e^{2x}.$$

Therefore,

$$\int x e^{2x} dx = \frac{x}{2} e^{2x} - \frac{1}{2} \int e^{2x} dx.$$

Thus,

$$= \frac{x}{2} e^{2x} - \frac{1}{4} e^{2x} + C.$$

Therefore,

$$\boxed{\int x e^{2x} dx = \frac{e^{2x}}{4} (2x - 1) + C}.$$

(d)

(i)

For an integral of the form

$$\int 5x^4 f(x^5) dx,$$

we would use **substitution**, because

$$\frac{d}{dx}(x^5) = 5x^4.$$

Thus the derivative of the inner function already appears in the integrand.

(ii)

For

$$\int x^2 \log(x) dx,$$

we would use **integration by parts**. A natural choice is

$$u = \log(x),$$

because differentiating $\log(x)$ simplifies it to $1/x$, while the polynomial x^2 is easy to integrate.

Problem 6: Improper and Multiple Integrals

(a)

$$\int_1^\infty \left(\frac{1}{3x}\right)^2 dx.$$

First simplify the integrand:

$$\left(\frac{1}{3x}\right)^2 = \frac{1}{9x^2}.$$

Thus,

$$\int_1^\infty \left(\frac{1}{3x}\right)^2 dx = \lim_{b \rightarrow \infty} \int_1^b \frac{1}{9x^2} dx.$$

Rewrite using powers:

$$= \frac{1}{9} \lim_{b \rightarrow \infty} \int_1^b x^{-2} dx.$$

An antiderivative is

$$\int x^{-2} dx = -x^{-1}.$$

Therefore,

$$= \frac{1}{9} \lim_{b \rightarrow \infty} \left[-\frac{1}{x} \right]_1^b.$$

Thus,

$$= \frac{1}{9} \lim_{b \rightarrow \infty} \left(-\frac{1}{b} + 1 \right).$$

Since

$$\lim_{b \rightarrow \infty} \frac{1}{b} = 0,$$

we obtain

$$\boxed{\frac{1}{9}}.$$

Therefore, the improper integral is

$$\boxed{\text{convergent}}.$$

(b)

$$\int_0^{\infty} e^{-x} dx.$$

Write the improper integral as a limit:

$$\int_0^{\infty} e^{-x} dx = \lim_{b \rightarrow \infty} \int_0^b e^{-x} dx.$$

An antiderivative is

$$-e^{-x}.$$

Therefore,

$$= \lim_{b \rightarrow \infty} \left[-e^{-x} \right]_0^b.$$

Thus,

$$= \lim_{b \rightarrow \infty} \left(-e^{-b} + 1 \right).$$

Since

$$e^{-b} \rightarrow 0 \quad \text{as} \quad b \rightarrow \infty,$$

we obtain

$$\boxed{1}.$$

Therefore, the integral is

$$\boxed{\text{convergent}}.$$

(c)

$$\int_{-\infty}^0 x^3 dx.$$

Write the improper integral as a limit:

$$\int_{-\infty}^0 x^3 dx = \lim_{a \rightarrow -\infty} \int_a^0 x^3 dx.$$

An antiderivative is

$$\frac{x^4}{4}.$$

Therefore,

$$= \lim_{a \rightarrow -\infty} \left[\frac{x^4}{4} \right]_a^0.$$

Thus,

$$= \lim_{a \rightarrow -\infty} \left(0 - \frac{a^4}{4} \right).$$

Since

$$a^4 \rightarrow \infty \quad \text{as} \quad a \rightarrow -\infty,$$

we have

$$-\frac{a^4}{4} \rightarrow -\infty.$$

Therefore,

$$\boxed{\text{divergent}}.$$

(d)

$$\int_0^1 \int_2^3 x^2 y^3 dx dy.$$

We first integrate with respect to x :

$$\int_2^3 x^2 y^3 dx = y^3 \int_2^3 x^2 dx.$$

Thus,

$$= y^3 \left[\frac{x^3}{3} \right]_2^3.$$

Therefore,

$$= y^3 \left(\frac{27}{3} - \frac{8}{3} \right) = \frac{19}{3} y^3.$$

Now integrate with respect to y :

$$\int_0^1 \frac{19}{3} y^3 dy = \frac{19}{3} \left[\frac{y^4}{4} \right]_0^1.$$

Thus,

$$= \frac{19}{3} \cdot \frac{1}{4} = \boxed{\frac{19}{12}}.$$

The order of integration is **not forced**, because all of the bounds are constants. The region is the rectangle

$$2 \leq x \leq 3, \quad 0 \leq y \leq 1.$$

Therefore, we could reverse the order and write

$$\int_2^3 \int_0^1 x^2 y^3 dy dx.$$

(e)

$$\int_0^1 \int_0^{\sqrt{1-x^2}} 2x^3 y dy dx.$$

First integrate with respect to y . Since x is constant with respect to y ,

$$\int_0^{\sqrt{1-x^2}} 2x^3 y dy = 2x^3 \left[\frac{y^2}{2} \right]_0^{\sqrt{1-x^2}}.$$

Therefore,

$$= x^3 \left[y^2 \right]_0^{\sqrt{1-x^2}}.$$

Thus,

$$= x^3(1 - x^2).$$

The outer integral becomes

$$\int_0^1 x^3(1 - x^2) dx = \int_0^1 (x^3 - x^5) dx.$$

An antiderivative is

$$\frac{x^4}{4} - \frac{x^6}{6}.$$

Therefore,

$$= \left[\frac{x^4}{4} - \frac{x^6}{6} \right]_0^1.$$

Thus,

$$= \frac{1}{4} - \frac{1}{6} = \frac{3-2}{12}.$$

Hence,

$$\boxed{\frac{1}{12}}.$$

With the bounds as written, the order is **forced**: we must integrate with respect to y first because the upper bound for y ,

$$\sqrt{1-x^2},$$

depends on x .

Geometrically, however, the region is the quarter unit disk

$$x^2 + y^2 \leq 1, \quad x \geq 0, \quad y \geq 0.$$

If we rewrite the bounds, we can reverse the order as

$$\int_0^1 \int_0^{\sqrt{1-y^2}} 2x^3 y dx dy.$$

Problem 7: Interpretation

(a) Reading a Partial Derivative

$$\boxed{\frac{\partial Approval_i}{\partial Employ_i} = 0.5}.$$

Holding Gas_i constant, a one-unit increase in the employment rate is associated with a 0.5-unit increase in predicted presidential approval. Holding gas prices fixed matters because it isolates the partial relationship between employment and approval rather than mixing it with changes in gas prices.

(b) Gradient versus Hessian

The condition $\nabla f(a) = 0$ is necessary for an interior differentiable local extremum, but it is not sufficient because a saddle point can also have zero gradient. For $g(x, y) = x^4 + y^4 - 4xy + 2$, the Hessian is indefinite at $(0, 0)$, so it is a saddle point, while the Hessian is positive definite at $(1, 1)$, so it is a local minimum. If $AC - B^2 = 0$, the second derivative test is inconclusive because the Hessian does not provide enough curvature information to classify the critical point.

(c) Integrals and Probability

The condition

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

means that the total probability over all possible values of the random variable is one. For an interval $[a, b]$,

$$\int_a^b f(x) dx = P(a \leq X \leq b),$$

so the area under the density curve between a and b is the probability that the random variable falls in that interval.